TEACHING WALLAH

Teaching Exam

Subject – Computer Science

Topic – Practice and Revision

Lecture No.- 01

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Topics to be Covered

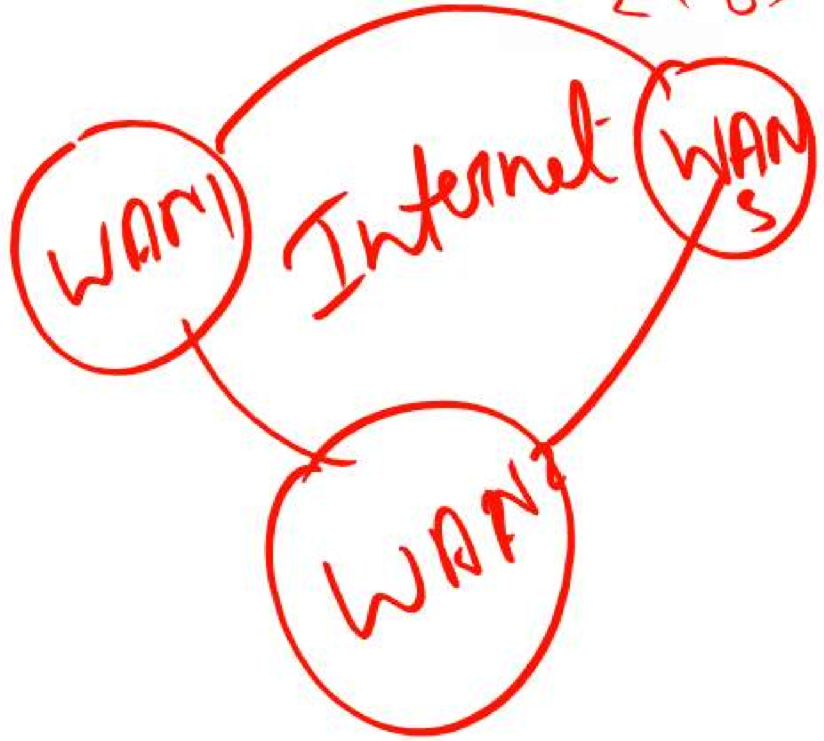
Topic

1:- Practice & Revision Set

1.What is a computer network?
a) A device used to display information on a computer screen
b) A collection of interconnected computers and devices that can communicate and share resources
c) A type of software used to create documents and presentations
d) The physical casing that protects a computer's internal components

Kajal, Rahul
1 (b)

2. What is internet?
a) A network of interconnected local area networks
b) A collection of unrelated computers
c) Interconnection of wide area networks
d) A single network



Answer 1: b

Explanation: A computer network refers to a collection of computers and devices linked together to share information, resources, and services. This interconnection enables communication, data sharing, and collaboration among the devices within the network.

Answer 2: c

Explanation: The internet is a global network formed by connecting wide area networks (WANs), enabling worldwide communication and data sharing.

3. Which of the following is an example of Bluetooth?

a) wide area network

b) virtual private network

c) local area network

d) personal area network

Logical Rul

Basic Network

4. Which of the following computer networks is built on the top of another network?

a) overlay network

b) prime network

c) prior network

d) chief network

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d) chief network

Topic- Practice

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Answer 3: d

Explanation: Bluetooth is a wireless technology used to create a wireless personal area network for data transfer up to a distance of 10 meters. It operates on 2.45 GHz frequency band for transmission.

Answer 4: ~~d~~ (a) Overlay Network → when we create Any logical Network Above the physical Network

Explanation: Bluetooth is a wireless technology used to create a wireless personal area network for data transfer up to a distance of 10 meters. It operates on 2.45 GHz frequency band for transmission.

5. What is the full form of OSI?
a) optical service implementation
b) open service Internet
c) open system interconnection → OSI Model
d) operating system interface

6. When a collection of various computers appears as a single coherent system to its clients, what is this called?
a) mail system
b) networking system
c) computer network
d) distributed system

Answer 5: c

Explanation: OSI is the abbreviation for Open System Interconnection. OSI model provides a structured plan on how applications communicate over a network, which also helps us to have a structured plan for troubleshooting. It is recognized by the ISO as the generalized model for computer network i.e. it can be modified to design any kind of computer network.

Answer 6: d

Explanation: A Computer network is defined as a collection of interconnected computers which uses a single technology for connection.

A distributed system is also the same as computer network but the main difference is that the whole collection of computers appears to its users as a single coherent system.

Example:- World wide web

7. How many layers are there in the ISO OSI reference model?
a) 7
b) 5
c) 4
d) 6

8. What are nodes in a computer network?
a) the computer that routes the data
b) the computer that terminates the data
c) the computer that originates the data
d) all of the mentioned

Answer 7: a

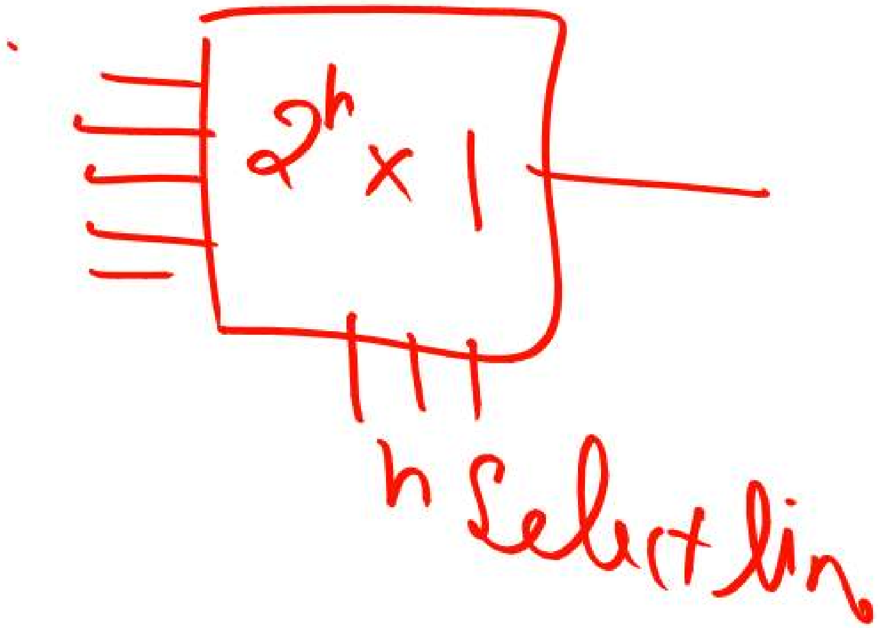
Explanation: The seven layers in ISO OSI reference model is Application, Presentation, Session, Transport, Network, Data link and Physical layer. OSI stands for Open System Interconnect and it is a generalized model.

Answer 8: d

Explanation: In a computer network, a node can be anything that is capable of sending data or receiving data or even routing the data to its destination. Routers, Computers and Smartphones are some examples of network nodes.

9. Which one of the following is not a function of network layer?
a) congestion control
b) error control
c) routing
d) inter-networking

10. How is a single channel shared by multiple signals in a computer network?
a) multiplexing
b) phase modulation
c) analog modulation
d) digital modulation



Answer 9: b

Explanation: In the OSI model, network layer is the third layer and it provides data routing paths for network communications. Error control is a function of the data link layer and the transport layer.

Answer 10: a

Explanation: In communication and computer networks, the main goal is to share a scarce resource. This is done by multiplexing, where multiple analog or digital signals are combined into one signal over a shared medium. The multiple kinds of signals are designated by the transport layer which is the layer present on a higher level than the physical layer.

11. Which of the following devices forwards packets between networks by processing the routing information included in the packet?
a) firewall
b) bridge
c) hub
d) router

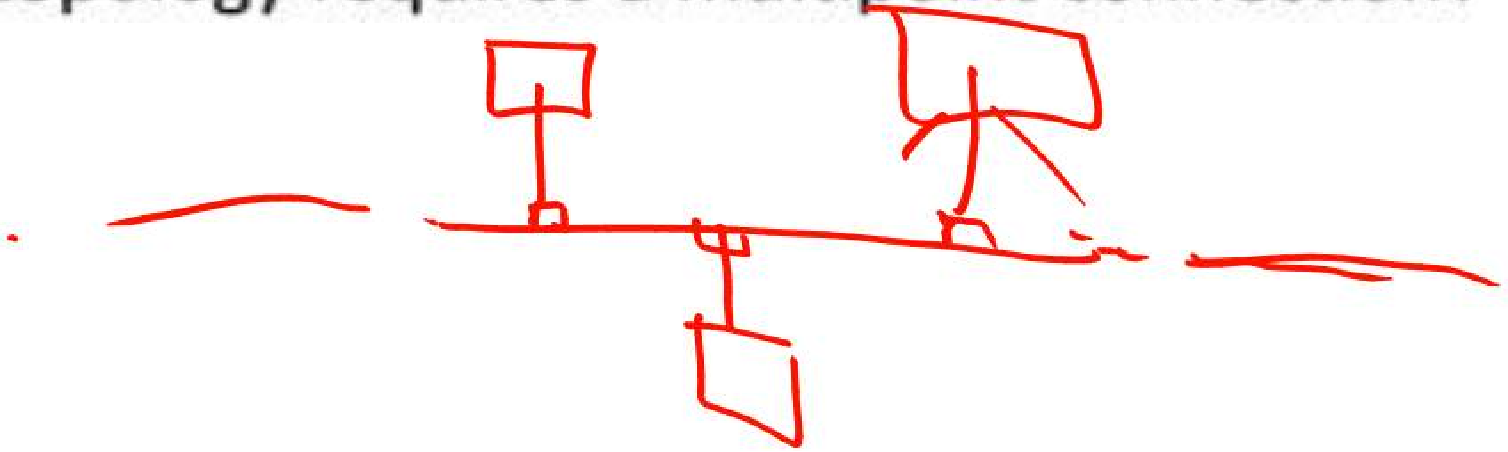
12. What is the term for an endpoint of an inter-process communication flow across a computer network?
a) port
b) machine
c) socket
d) pipe

Answer 11: d
Explanation: A router is a networking device that forwards data packets between computer networks. Routers perform the traffic directing functions on the Internet. They make use of routing protocols like RIP to find the cheapest path to the destination.

Answer 12: c
Explanation: Socket is one end point in a two way communication link in the network. TCP layer can identify the application that data is destined to be sent by using the port number that is bound to socket.

13. Which of this is not a network edge device?
a) Switch
b) PC
c) Smartphones
d) Servers

14. Which topology requires a multipoint connection?
a) Ring
b) Bus
c) Star
d) Mesh



15. Which of the following maintains the Domain Name System?

DNS

a) a single server
b) a single computer
c) distributed database system
d) none of the mentioned

16. Which of the following are Gigabit Ethernets?
a) 1000 BASE-LX
b) 1000 BASE-CX
c) 1000 BASE-SX
d) All of the mentioned

Answer 15: c

Explanation: A domain name system is maintained by a distributed database system. It is a collection of multiple, logically interrelated databases distributed over a computer network.

E-Thernet
wire
Standard

Answer16: d

Explanation: In computer networking, Gigabit Ethernet (GbE or 1 GigE) is a term describing various technologies for transmitting Ethernet frames at a rate of a gigabit per second (1,000,000,000 bits per second), as defined by the IEEE 802.3-2008 standard. It came into use beginning in 1999, gradually supplanting Fast Ethernet in wired local networks, as a result of being considerably faster.

17. Which layer is responsible for process to process delivery in a general network model?
a) session layer
b) data link layer
c) transport layer → end to end Transmission → TCP Protocol → Data Packet Safely Reach
d) network layer
UDP ,, → Speed

18. What is the term for the data communication system within a building or campus?
a) MAN
b) LAN
c) PAN
d) WAN

Answer 17: c

Explanation: The role of Transport layer (Layer 4) is to establish a logical end to end connection between two systems in a network. The protocols used in Transport layer is TCP and UDP. The transport layer is responsible for segmentation of the data. It uses ports for the implementation of process-to-process delivery.

Answer18: b

Explanation: LAN is an abbreviation for Local Area Network. This network interconnects computers in a small area such as schools, offices, residence etc. It is the most versatile kind of data communication system where most of the computer network concepts can be visibly used.

19. What was the name of the first network?
a) ASAPNET
b) ARPANET
c) CNNET
d) NSFNET

20. Which of the following is the network layer protocol for the internet?
a) hypertext transfer protocol
b) file transfer protocol
c) ethernet
d) internet protocol

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Answer 19: b
Explanation: ARPANET stands for Advanced Research Projects Agency Networks. It was the first network to be implemented which used the TCP/IP protocol in the year 1969.

TCP/IP

Answer 20: d
Explanation: There are several protocols used in Network layer. Some of them are IP, ICMP, CLNP, ARP, IPX, HRSP etc. Hypertext transfer protocol is for application layer and ethernet protocol is for data link layer.

21. If a link transmits 4000 frames per second, and each slot has 8 bits, what is the transmission rate of the circuit using Time Division Multiplexing (TDM)?

a) 500kbps

b) 32kbps

c) 32bps

d) 500bps

→ Digital Signal

1 frame = 8 bit

4000 frame = 8×4000 bits

= 32000 bits

= 32×10^3 bits ⇒ 32kbps

22. Which of the following allows LAN users to share computer programs and data?

a) File server

b) Network

c) Communication server

d) Print server

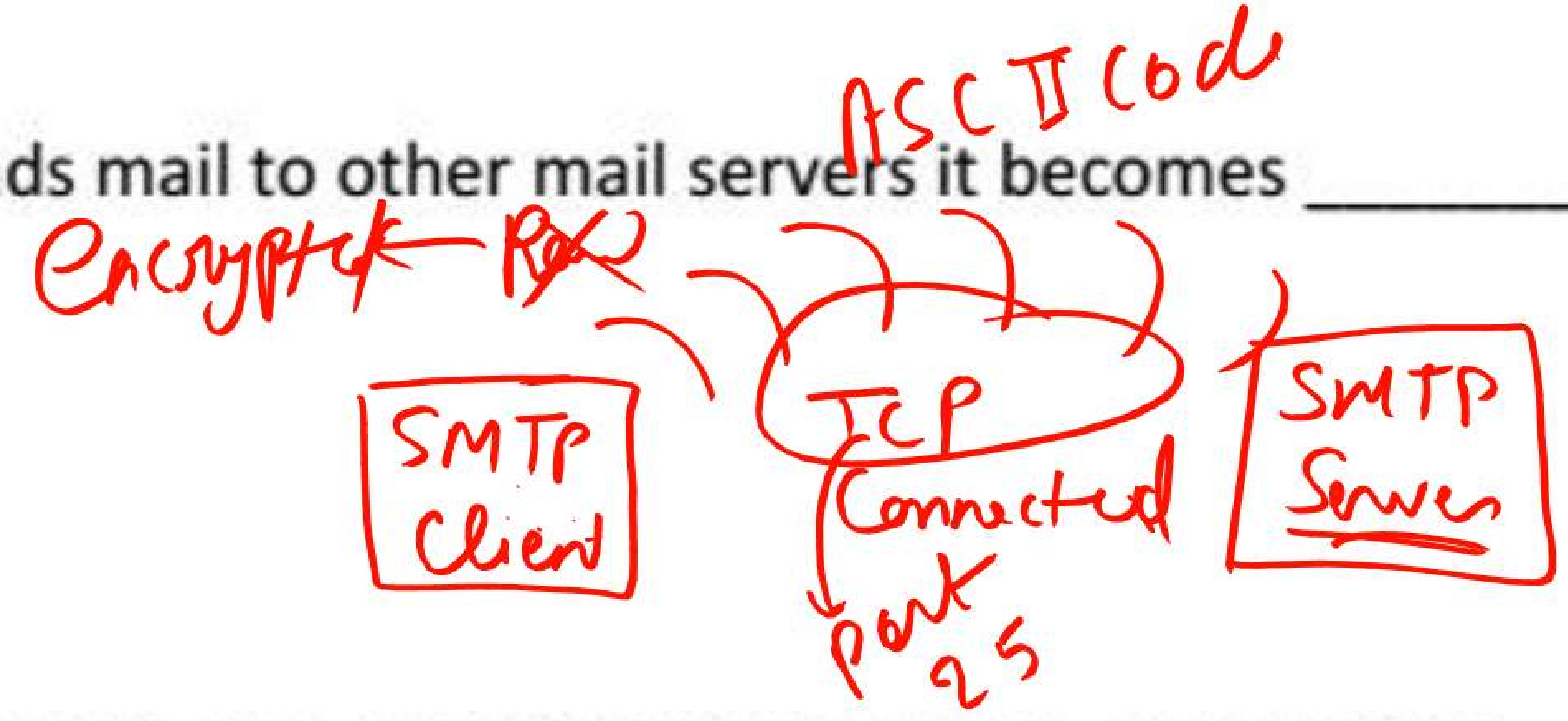
→ FTP (file transmit from one device to Another)

Answer 21: b
Explanation: Transmission rate= frame rate * number of bits in a slot.
Given: Frame rate = 4000/sec and number of bits in slot = 8
Thus, Transmission rate = (4000 * 8) bps
= 32000bps
= 32kbps

FTPS → SSL/TLS

Answer 22: a
Explanation: A file server allows LAN users to share computer programs and data. It uses the File Transfer Protocol to provide this feature on ports 20 and 21. The file server works as a medium for the transfer.

23. When the mail server sends mail to other mail servers it becomes ______________
a) SMTP server
b) SMTP client
c) Peer
d) Master



24. If you have to send multimedia data over SMTP it has to be encoded into _______
a) Binary
b) Signal
c) ASCII → Code → 7 bit encrypted bit
d) Hash

Answer 23: b

Explanation: SMTP clients are the entities that send mails to other mail servers. The SMTP servers cannot send independent mails to other SMTP servers as an SMTP server. There are no masters or peers in SMTP as it is based on the client-server architecture.

Answer 24: c

Explanation: Since only 7-bit ASCII codes are transmitted through SMTP, it is mandatory to convert binary multimedia data to 7-bit ASCII before it is sent using SMTP.

25. Expansion of SMTP is ________
a) Simple Mail Transfer Protocol
b) Simple Message Transfer Protocol
c) Simple Mail Transmission Protocol
d) Simple Message Transmission Protocol

26. The underlying Transport layer protocol used by SMTP is ________
a) TCP
b) UDP
c) Either TCP or UDP
d) IMAP

Answer 25: a

Explanation: SMTP or Simple Mail Transfer Protocol is an application layer protocol used to transport e-mails over the Internet. Only 7-bit ASCII codes can be sent using SMTP.

Answer 26: a

Explanation: TCP is a reliable protocol, and Reliability is a mandatory requirement in e-mail transmission using SMTP.

27. Choose the statement which is wrong incase of SMTP?

a) It requires message to be in 7bit ASCII format

b) It is a pu~~ll protocol~~ Push protocol

c) It transfers files from one mail server to another mail server

d) SMTP is responsible for the transmission of the mail through the internet

28. Typically the TCP port used by SMTP is __________

a) 25

b) 35

c) 50

d) 15

HTTP, POP3, IMAP

Pull Protocol

Response

Inbox

Device download

Pull (Mail Server)

(Real Time)

Answer 27: b

Explanation: In SMTP, the sending mail server pushes the mail to receiving mail server hence it is push protocol. In a pull protocol such as HTTP, the receiver pulls the resource from the sending server.

Answer 28: a

Explanation: The ports 15, 35 and 50 are all UDP ports and SMTP only uses TCP port 25 for reliability.

UDP Ports → 15, 35, 50

TCP ,, → 25

29. A session may include _________
a) Zero or more SMTP transactions
b) Exactly one SMTP transactions
c) Always more than one SMTP transactions
d) Number of SMTP transactions cant be determined

30. Which of the following is an example of user agents for e-mail?
a) Microsoft Outlook
b) Facebook
c) Google
d) TumblrAnswer

Gmail app

29. Answer: a

Explanation: An SMTP session consists of SMTP transactions only even if no transactions have been performed. But no transactions in the session might mean that the session is inactive or is just initiated.

30. Answer: a

Explanation: Among the options, only Microsoft Outlook is an e-mail agent. Google is a search engine and Facebook, and Tumblr are social networking platforms. Gmail and Alpine are some other examples of e-mail agent.

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THANK YOU